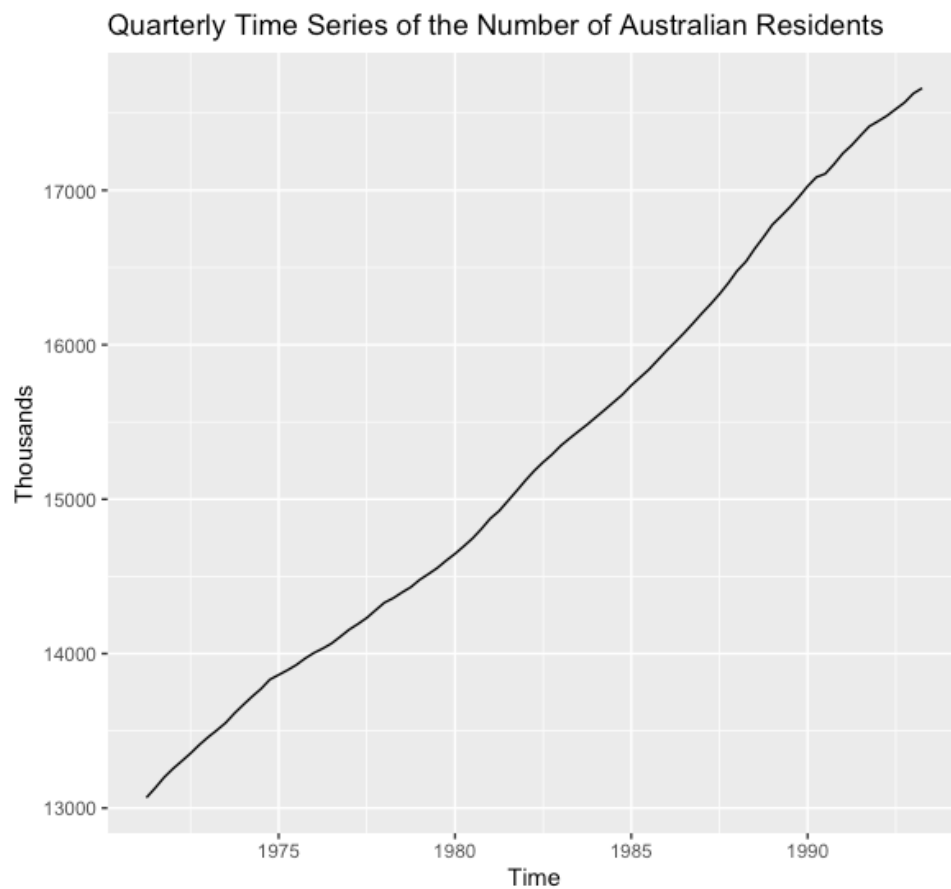


Question 1

Consider the numbers (in thousands) of Australian residents measured quarterly from March 1971 to March 1994 in the time series plot below:



Suppose you decide to fit a Holt's linear trend exponential smoothing model to forecast future values for this time series.

- Comment on why such a model would be suitable to forecast future values.
- What simple forecasting method would serve as a useful benchmark for this time series?

Consider the **subset of data**, from March 1992 to March 1993 below, with the output from fitting the exponential smoothing model and a simple forecasting model on the entire time series (i.e. from March 1971 to March 1994).

Year	Quarter	t	Yt	lt	bt	Y^t
1991	Mar	81	17292.00	17292.00	56.90	17299.71
	Jun	82	17354.20	17354.20	59.24	C
	Sep	83	17414.20	A	59.58	17413.44
	Dec	84	17447.30	17447.30	47.87	17473.78
1992	Mar	85	17482.60	17482.60	42.31	17495.18
	Jun	86	17526.00	17526.00	B	17524.92
	Sep	87	17568.70	17568.70	42.75	17568.79

	Dec	88	17627.10	17627.10	49.67	D
1993	Mar	89	17661.50	17661.50	42.92	17676.77
	Jun	1				17704.42
	Sep	2				E
	Dec	3				F

Forecast method: Holt's method

Model Information:

Holt's method

Call:

```
holt(y = austres, h = 4)
```

Smoothing parameters:

alpha = 0.9999

beta = 0.4421

Initial states:

l = 13006.2854

b = 77.9696

sigma: 10.4239

Forecast method: Random walk with drift

Model Information:

Call: rwf(y = austres, h = 4, drift = TRUE)

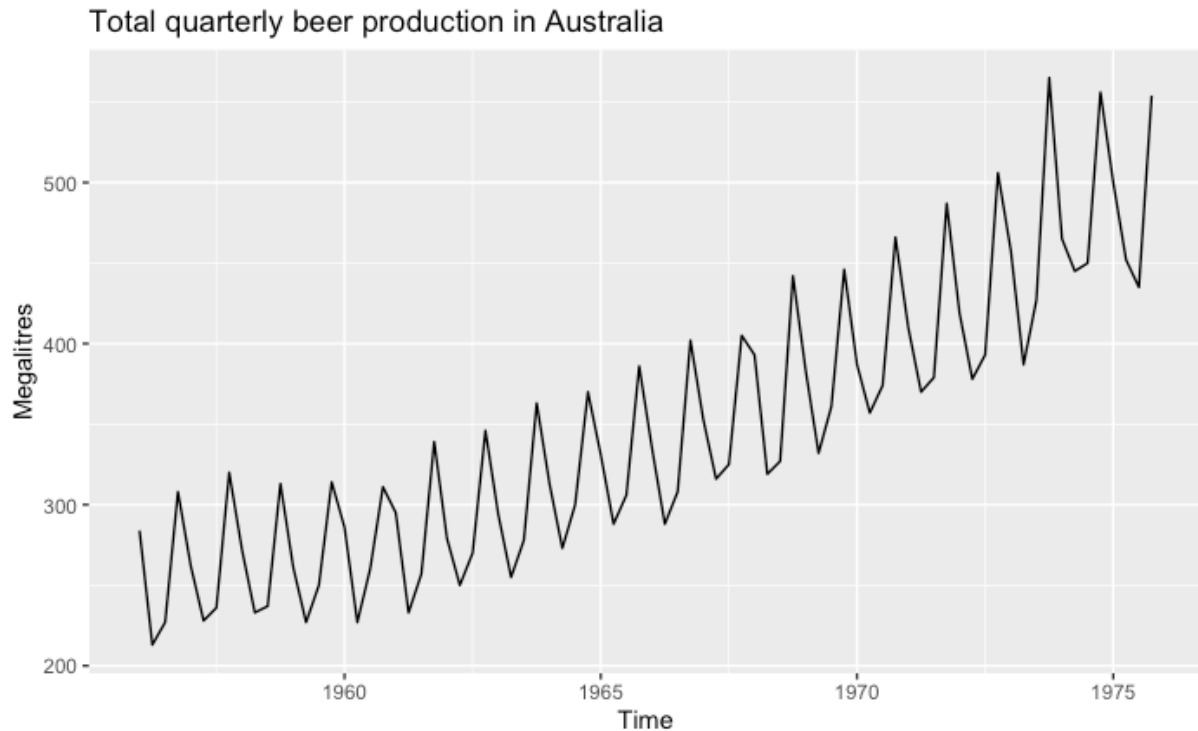
Drift: 52.2068 (se 1.3533)

Residual sd: 12.6951

- Compute the values A-F in the table
- When calculating the level in any given time period, how much weight is placed on the most recent/current level information? Interpret this value.
- When calculating the slope in any given time period, how much weight is placed previous slope information? Interpret this value.
- Forecast the quarterly values for Jun, Sep and Dec 1993 with the Drift model
- If the MAD for the Drift model is 10.26 and for the Holts model is 7.51, what can you conclude?

Question 2

Consider the total quarterly beer production in Australia, from 1956 to 1975



- What would be the most appropriate simple forecasting method for this time series for short-term forecasts?
- If our time series only consisted of the data from 1956 to 1960, explain why the simple forecasting method in (a) would produce increasingly inaccurate forecasts in the future?

A Holt-Winters Exponential smoothing model was fit to the data from 1968 to 1971. Consider the output from the fitted model, and answer the questions that follow.

Parameters

alpha	0.028
beta	0.0001
gamma	0.0001

Quarter	t	Y _t	I _t	b _t	S _t	Y [^] _t
Q1	-3				1.0207	
Q2	-2				0.8901	
Q3	-1				0.9230	

Q4	0		365.3337	3.0193	1.1662	
Q1 - 1968	1	393	368.8200	3.0193	1.0207	375.9779
Q2 - 1968	2	319	371.4626	3.0193	0.8901	D
Q3 - 1968	3	327	A	3.0192	0.9230	345.6468
Q4 - 1968	4	442	376.9935	3.0192	1.1662	439.5821
Q1 - 1969	5	383	379.8789	3.0192	1.0207	387.8790
Q2 - 1969	6	332	382.6207	3.0192	0.8901	340.8176
Q3 - 1969	7	361	385.7932	B	0.9230	355.9456
Q4 - 1969	8	446	388.6339	3.0192	1.1662	453.4330
Q1 - 1970	9	387	391.3031	3.0192	1.0207	399.7603
Q2 - 1970	10	357	394.5115	3.0192	0.8901	350.9863
Q3 - 1970	11	374	397.7455	3.0192	0.9230	366.9208
Q4 - 1970	12	466	400.7318	3.0192	C	467.3718
Q1 - 1971	13	410	403.6932	3.0192	1.0207	412.1086
Q2 - 1971	14	370	406.9636	3.0192	0.8901	362.0147
Q3 - 1971	15	379	410.0006	3.0192	0.9230	378.4141
Q4 - 1971	16	487	413.1479	3.0192	1.1662	481.6637
Q1 - 1972	1	419				E
Q2 - 1972	2	378				F
Q3 - 1972	3	393				G
Q4 - 1972	4	506				H

- c) Compute values A – H (Use the equations on the formulae sheet). Round to 3 decimal places.
- d) Compute the MSE value for this model for the values in the test set.